# Differentiating the effect of material and junction resistances in nano-networks

Cian Gabbett,[1*] Adam G. Kelly,[1*] Emmet Coleman,[1] Luke Doolan,[1] Tian Carey,[1] Kevin Synnatschke,[1] Shixin Liu,[1] Anthony Dawson,[1] Domhnall O'Suilleabhain,[1] Jose Munuera,[1] Eoin Caffrey,[1] John B. Boland,[1] Zdeněk Sofer,[2] Goutam Ghosh,[3] Sachin Kinge,[4] Laurens D.A. Siebbeles,[3] Neelam Yadav,[5] Jagdish K. Vij,[5] Muhammad Awais Aslam,[6] Aleksandar Matkovic[6] and Jonathan N. Coleman[1]**

[1]*School of Physics, CRANN & AMBER Research Centres, Trinity College Dublin, Dublin 2, Ireland*

[2]*Department of Inorganic Chemistry, University of Chemistry and Technology Prague, Technická 5, Prague 6, 166 28, Czech Republic*

[3]*Chemical Engineering Department, Delft University of Technology, Van der Maasweg 9, NL-2629 HZ Delft, The Netherlands*

[4]*Materials Research & Development, Toyota Motor Europe, B1930 Zaventem, Belgium*

[5]*Department of Electronic & Electrical Engineering, Trinity College Dublin 2, Ireland*

[6]*Chair of Physics, Department Physics, Mechanics and Electrical Engineering, Montanuniversität Leoben, Franz Josef Strasse 18, 8700 Leoben, Austria*

*These authors contributed equally, **colemaj@tcd.ie

ABSTRACT: Networks of nanowires and nanosheets are important for many applications in printed electronics. However, network conductivity and mobility are usually limited by the inter-particle junction resistance. Minimising junction resistance is challenging because it is difficult to measure. Here we develop a simple model for conduction in networks of 1D or 2D nanomaterials, which allows us to extract junction and nanoparticle resistances from particle-size-dependent D.C. resistivity data. We find junction resistances in porous networks to scale with nanoparticle resistivity and vary from 5Ω for silver-nanosheets to 25 GΩ for $WS_2$-nanosheets. Moreover, our model allows junction and nanoparticle resistances to be obtained from A.C. impedance spectra of semiconducting networks. Impedance data links the high-mobility (~7 $cm^2/Vs$) of aligned networks of electrochemically-exfoliated $MoS_2$ nanosheets to low junction resistances of ~670 kΩ. Temperature-dependent impedance measurements allow us to quantitatively differentiate intra-nanosheet phonon-limited bandlike transport from inter-nanosheet hopping for the first time.

Printed electronic (PE) devices are increasingly important due to their flexibility, scalability, and cost-effectiveness.[1] Driven by their combination of solution-processibility and reasonable electrical performance, 0D, 1D, and 2D nanoparticles have now been widely explored as PE materials.[1,2] Printed networks of carbon nanotubes have shown great promise for use in transistors,[3,4] LEDs[5] and photodetectors[6] while metallic nanowire networks have been heavily studied as transparent electrodes,[7,8] EMI shields[9] and heating elements.[10] More recently, solution-processed networks of 2D materials such as graphene and $MoS_2$ have been investigated for a broad range of applications in all areas of (opto)electronics and energy storage.[11-13]

The success of this approach relies on exploiting the exceptional intrinsic properties of the individual nanoparticles (e.g. high conductivity or carrier mobility), when they are assembled into large-area networks. Conductive nanoparticles have been printed into networks with conductivities reaching ~$10^5$ S $m^{-1}$ for graphene,[12] ~$10^6$ $Sm^{-1}$ for MXenes,[14] >$10^6$ $Sm^{-1}$ for silver nanowires (AgNWs)[7,15] and >$10^7$ $Sm^{-1}$ for silver nanosheets (AgNS).[16] While the properties of conducting networks now approach those of the constituent nanoparticles, semiconducting networks have not kept up. Semiconducting carbon nanotube networks display mobilities <100 $cm^2$ $V^{-1}$ $s^{-1}$ [ref. [18]] compared to >$10^5$ $cm^2$ $V^{-1}$ $s^{-1}$ for an individual nanotube[17], while networks of solution-processed $MoS_2$ achieve ~10 $cm^2$ $V^{-1}$ $s^{-1}$ [refs. [12,18-20]] compared to >100 $cm^2$ $V^{-1}$ $s^{-1}$ for mechanically exfoliated nanosheets.[21-23] The reason for these discrepancies is simple: almost all networks are limited by the junctions between particles, with semiconducting networks particularly junction-limited.[12,24]

Realising high-performance printed devices requires minimizing the junction resistance ($R_J$) relative to the nanoparticle resistance ($R_{NP}$). Achieving $R_J << R_{NP}$ would yield networks with properties approaching individual nanoparticles. Strategies to achieve this include optimising nanoparticle dimensions or deposition techniques[12] and chemical cross-linking.[25,26] However, due to our inability to easily measure either the junction or the nanoparticle resistance *in-situ*, assessing the progress of various strategies towards achieving $R_J < R_{NP}$ is difficult.

Despite their importance, the literature contains very little quantitative data on junction resistances. Although conductive-AFM[27] or micro-electrodes[28] can yield local information on both nanoparticle and junction resistances, these methods are unsuitable for large-area printed networks. Alternatively, although models exist linking junction resistance to network conductivity they are too complex for routine utilisation.[29,30] The resultant lack of basic

information has hindered printed device development and forced a reliance on trial-and-error for device optimisation.

Here, we develop a simple model relating the conductivity and mobility of nanoparticle networks to controllable nanoparticle parameters and network properties including junction resistance. We show that this model accurately describes experimental data for various nanomaterials and allows the extraction of both nanoparticle and junction resistances. We combine this model with impedance spectroscopy measurements to develop a powerful technique for simultaneously measuring both nanosheet and junction resistances within networks of semiconducting nanosheets.

***Model Development***

We utilise a classical, circuit-based approach to derive an equation for the resistivity of networks of 1D or 2D nanoparticles (e.g. nanowires or nanosheets), $\rho_{Net}$, in terms of properties of individual nanoparticles as well as the junction resistance, $R_J$, and network porosity, $P_{Net}$ (see SI S2 for derivation). Assuming a carrier traverses a well-defined current path (Fig. 1a) consisting of a linear array of nanoparticles (Fig. 1b), it must cross an inter-particle junction every time it traverses a nanoparticle. Current paths can then be modelled as linear arrays of nanoparticle-junction pairs, each pair represented by two resistors (Fig. 1b), describing the average nanoparticle ($R_{NP}$) and junction ($R_J$) resistances.

By relating the number of resistor pairs in a path to the channel length and the average distance travelled within each nanoparticle, one can estimate typical voltage drops across individual nanoparticles and junctions upon application of a voltage. These voltage drops yield the average transit times across individual nanoparticles and junctions. Combining these equations with an expression for the total transit time through the channel, one can obtain an equation for the network mobility:

$$\mu_{Net} \approx \frac{\mu_{NP}}{\left[1+\frac{R_J}{R_{NP}}\right]\left[1+\frac{2}{n_{NP} l_{NP} A_{NP}}\right]} \qquad (1)$$

where $\mu_{NP}$, $n_{NP}$, $l_{NP}$ and $A_{NP}$ are nanoparticle mobility, carrier density, length and cross-sectional area. This equation shows clearly that $\mu_{Net}$ depends on $R_J / R_{NP}$, which should be small to maximise mobility. Then combining an expression for network resistivity[12] (

$\rho_{Net}^{-1} = (1-P_{Net})n_{NP}e\mu_{Net}$) with expressions relating $A_{NP}$ and $R_{NP}$ to nanoparticle geometry, we can generate equations for $\rho_{Net}$ specific to 1D nanowires/nanotubes and 2D nanosheets:

$$\rho_{Net} \approx \frac{1}{(1-P_{Net})}\left[\rho_{NW} + \frac{\pi D_{NW}^2 R_J}{2l_{NW}}\right]\left[1+\frac{8}{n_{NW}l_{NW}\pi D_{NW}^2}\right] \quad \text{(1D, 2a)}$$

$$\rho_{Net} \approx \frac{\left[\rho_{NS} + 2t_{NS}R_J\right]}{(1-P_{Net})}\left[1+\frac{2}{n_{NS}t_{NS}l_{NS}^2}\right] \quad \text{(2D, 2b)}$$

where $D_{NW}$ and $t_{NS}$ are the nanowire diameter and nanosheet thickness and the geometry-specific subscripts *NS* and *NW* refer to nanosheet and nanowire, respectively (see SI S2 for 0D expressions). For large values of $n_{NP}$ such as those found for graphene, AgNS or AgNWs, the second square-bracketed terms can be ignored.

***Measuring the change in network resistivity with nanoparticle dimensions***

Equations (2a) and (2b) suggest a rich array of size-dependent behaviour that has not yet been observed (see SI S2). For example, the appearance of nanosheet size parameters ($t_{NS}$, $l_{NS}$) in both denominator and numerator predicts a non-monotonic size-dependence with either a positive or negative $d\rho_{Net}/dt_{NS}$. To search for such behavior and to test the validity of equations (2a) and (2b), we produced inks of 1D AgNWs and four types of 2D nanosheets; graphene, $WS_2$, and $WSe_2$ (synthesised by liquid-phase exfoliation[31]) and commercial AgNS (see Fig. 2 and SI Methods and Materials for detailed methods). Each material was size-selected into fractions (Fig. 2a, c, d) which were spray-coated to produce a set of networks for electrical testing (see Figs. 2b, e-h for representative SEM images).

The measured size-dependent D.C. resistivity is shown in Figures 3a-e for all five materials. Because $n_{NW}$ is large for AgNWs, equation (2a) predicts that $\rho_{Net}$ scales linearly with $(l_{NW})^{-1}$, behaviour that is clearly seen in Figure 3a. Nanosheets produced by LPE[32] and the AgNS (see SI) display a roughly constant aspect ratio, $k_{NS}$, allowing us to reduce the number of variables in equation (2b) by replacing $t_{NS}$ with $t_{NS} = l_{NS}/k_{NS}$. Ignoring the final term in equation (2b) for graphene and AgNS, we now find that $\rho_{Net}$ should scale linearly with $l_{NS}$, as seen experimentally in Figures 3b-c. However, for semiconducting materials, $n_{NS}$ is low meaning the final term in equation (2b) must be considered and which predicts a resistivity-minimum at a specific nanosheet size. Figures 3d-e show $\rho_{Net}$ for $WS_2$ and $WSe_2$ which initially falls with increasing $l_{NS}$ before reaching a minimum, behaviour which is consistent with our non-intuitive

prediction. That such a minima exists is important as it suggests the existence of an optimal nanosheet size.

Equations (2a) and (2b) describe our data well, including the counter-intuitive trends, with fitting yielding values for $R_J$ and $R_{NP}$ as shown in the figure panels (see SI S3). Converting $R_{NP}$ to $\rho_{NP}$ (see SI S2) yields $\rho_{NW}$=1.5±0.2×10$^{-8}$ Ω m (AgNWs) and $\rho_{NS}$=7.2±3.9×10$^{-8}$ Ω m (AgNSs), similar to bulk silver (1.6×10$^{-8}$ Ω m). For graphene $\rho_{NS}$=1.7±0.5×10$^{-5}$ Ω m, consistent with in-plane graphite (~10$^{-5}$-10$^{-6}$ Ω m)[33]. The $\rho_{NS}$ values for $WS_2$ (1.9±0.3 Ω m) and $WSe_2$ (0.35±0.07 Ω m) were consistent with previously-reported values (0.6 Ω m [ref.[34]] and 0.1 Ω m [ref. [35]], respectively). The values of $R_J$ ranged from 5.2±0.7 Ω for AgNS to 24±2.3 GΩ for $WS_2$ and are consistent with previous estimates of $R_J$ ~3 Ω for AgNS,[16] 185 Ω for AgNWs,[36] ~10$^3$-10$^5$ W for graphene[12] and ~10$^9$ Ω for $MoS_2$.[37] We note that $R_J$ scales with $\rho_{NP}$ (figure 3F) and that $R_J/R_{NP}$-values were all >1 indicating that these networks were predominately junction-limited.

***A direct measurement of $R_J$ and $R_{NP}$ using impedance spectroscopy***

Measuring $R_{NP}$ and $R_J$ as described above is time-consuming. We propose that A.C. impedance spectroscopy, a powerful tool for device and materials characterisation,[38,39] can leverage the inevitable junction capacitance to extract information about nanosheet and junction resistances. While similar proposals have been made previously for grain/grain-boundary[40,41,42] as well as 2D[43] systems, such measurements probe all junctions in all current paths, in practice yielding $R_{NP}$ and $R_J$ in arbitrary units (but *not* absolute values), limiting useful analysis.

To extract absolute values of $R_{NS}$ and $R_J$ for nanosheet networks, the impedance spectra of the network ($Z_{Net}$) must be converted to spectra representing the average nanosheet-junction pair ($Z_{NS\text{-}J}$) within the network. These NS-J spectra can then be analysed based on microscopic considerations (see below).

Equation (2b) relates the D.C. resistivity of a nanosheet network, $\rho_{Net}$, to $(R_{NS} + R_J)$, the resistance of the average nanosheet-junction pair. We propose the same scaling exists between the complex resistivity of the network, $\rho^*_{Net}$, and $Z_{NS\text{-}J}$ (where $\rho^*_{Net} = Z_{Net}A/L$, and $A$ and $L$ are the network cross-sectional area and channel length, see SI S9–S10). This yields an equation which converts the real and imaginary parts of $\rho^*_{Net}$ to those representing the nanosheet-junction pair, once $P_{Net}$, $t_{NS}$, $l_{NS}$ and $n_{NS}$ are known:

$$Z_{NS-J} = \rho^*_{Net} \frac{(1-P_{Net})}{2t_{NS}} \left[ 1 + \frac{2}{n_{NS} t_{NS} l_{NS}^2} \right]^{-1} \quad (3)$$

We demonstrate this impedance approach using liquid-deposited networks of electrochemically exfoliated (EE) $MoS_2$ nanosheets ($l_{NS} \approx 1$ μm, $t_{NS} \approx 3.3$ nm) with low porosity[44] and large-area junctions[18] (Fig. 4a). While this is an intensively studied system due to its relatively high mobility (>1 $cm^2 V^{-1} s^{-1}$ for printed networks)[18-20,45], the actual $R_{NS}$ and $R_J$ values are completely unknown. We first measure the (peak) field-effect mobility of these networks in a transistor geometry (Fig. 4b and SI S19), obtaining an average of $\mu_{Net}$=6.6±0.6 $cm^2 V^{-1} s^{-1}$, consistent with previous measurements.[18]

We then measured the real and imaginary parts of the complex resistivity as a function of frequency as shown in Fig. 4c (see SI S4-S17 for detailed information. Preliminary data for other materials is in SI S17). The low-frequency plateau of Re($\rho^*_{Net}$) yields a D.C. resistivity of $\rho_{Net}$ = 0.024 Ω m. Combining this value with the measured mobility gives a carrier density for the network of $3.8\times10^{23}$ $m^{-3}$, similar to previous values for EE-$MoS_2$.[20,46]

With these values now known, equation (3) can be used to convert the network impedance, $Z_{Net}$, into the impedance of the average nanosheet-junction pair, $Z_{NS-J}$. Figure 4d shows the real component of $Z_{NS-J}$, with the inset showing the imaginary component. In the A.C. domain, the nanosheet-junction pair can be described as the nanosheet resistance ($R_{NS}$) in series with a parallel resistor ($R_J$) and capacitor ($C_J$) representing the junction (Fig. 4c, inset), an arrangement referred to as the Randles circuit. The $Z_{NS-J}$ spectrum can be fitted using equations appropriate to the Randles circuit to yield values of $R_{NS}$, $R_J$ and $C_J$. We account for the inevitable distribution of junction resistances by fitting the data using a modified equation for the Randles circuit[47] (see SI S11-S12):

$$\mathrm{Re}\, Z_{NS-J}(\omega) = R_{NS} + \frac{R_J \left[ 1 + (\omega R_J C_J)^n \cos(n\pi/2) \right]}{1 + 2(\omega R_J C_J)^n \cos(n\pi/2) + (\omega R_J C_J)^{2n}} \quad (4)$$

where $n$ is an ideality factor that decreases from 1 as the distribution of $R_J C_J$ values in the network broadens.[48]

We find excellent fits yielding values of $R_J$=(2.9±0.1) MΩ, $R_{NS}$=(0.67±0.07) MΩ, $C_J$=(8.4±0.4)×$10^{-15}$ F and n≈0.985. Over five devices on the same substrate, $R_{NS}$ typically varies by <20% with $R_J$ and $C_J$ showing larger spreads (SD/mean<60%) due to morphology

variations. Here $R_J$ is >1000× lower than in Figures 3d-e for LPE nanosheets, while $R_J/R_{NS}$=4.4±0.2, meaning it is much less junction-limited than the LPE $WS_2$ and $WSe_2$ networks above. We can further analyse the nanosheet resistance by converting it to nanosheet resistivity (see SI S2), giving $\rho_{NS}$=(4.4±0.5)×$10^{-3}$ Ω m. As the meso-porosity of the network is extremely low (~0.02),[44] we assume the network carrier density is the same as the nanosheet carrier density which then implies a nanosheet mobility of 37±4 $cm^2$ $V^{-1}$ $s^{-1}$, reasonable for solution-processed $MoS_2$.[20,49]

We can support this result using several direct measurements. First, we used time-resolved pump-probe terahertz spectroscopy to determine the room-temperature A.C. mobility of photogenerated charge carriers. The observed mobility at a frequency of 1 THz is 40±2 $cm^2$ $V^{-1}$ $s^{-1}$ (see SI S18), consistent with the value implied by impedance. Second, we performed field-effect mobility measurements on individual $MoS_2$ nanosheets (see Fig. 4e and SI S20) obtaining a zero-gate-bias value of 42±6 $cm^2$ $V^{-1}$ $s^{-1}$, again consistent with our results. Combining this value with our $\mu_{Net}$-value and reformulating equation (1) as $R_J / R_{NS} \approx \mu_{NS} / \mu_{Net} - 1$, (neglecting the final term as $n_{NS}$ is large) allows us to estimate $R_J/R_{NS}$=5.3±1.4, again within error of the impedance result.

We then used *in-operando* frequency-modulated Kelvin Probe Force Microscopy (KPFM) measurements to map out the spatial distribution of the electrostatic potential across a network (Figs. 4f-g).[50-52] Between the biased and grounded electrodes, we find a combination of gradual decreases in potential within the nanosheets and well-defined potential drops at the junctions. By summing the potential drops at the junctions along the channel length, we extract the overall fraction of potential drop within nanosheets which yields a mean value of $R_J/R_{NS}$ =10±4 (Fig. 4h). Although microstructural variations in similarly deposited networks will cause differences in $R_J$, we find these data to be extremely consistent supporting the validity of the impedance method.

***Using the impedance method: temperature dependence***

Impedance spectroscopy allows $R_J$ and $R_{NS}$ to be measured simultaneously under various circumstances. We demonstrate this by performing impedance measurements on networks of electrochemically-exfoliated $MoS_2$ at various temperatures (Fig. 5). The low frequency limit of the Re($\rho^*_{Net}$) spectrum (Fig. 5a) yields the network resistivity ($\rho_{Net}$) which is plotted versus 1/T in Figure 5b. Previous measurements on electrochemically-exfoliated $MoS_2$ networks have

shown $\rho_{Net}$ to follow activated behaviour around room temperature ($\rho = \rho_0 \exp(E_a / kT)$, $\rho_0$ and $E_a$ are constants) but 3D variable-range hopping[53] at lower temperatures ($\rho = \rho_0 \exp[(T_0 / T)^{1/4}]$, $\rho_0$ and $T_0$ are constants).[54] As shown in Figure 5b and 5b (inset), our data is consistent with this behaviour (fit constants in panel). This standard analysis cannot distinguish the respective contributions from the nanosheets and junctions. To decouple these properties, we first convert the network impedance spectra to Re($Z_{NS-J}$) and Im($Z_{NS-J}$) spectra (Figures 5c-d), finding a well-defined temperature dependence.

Fitting the Re($Z_{NS-J}$) spectrum to equation (4) yields values of $R_{NS}$, $R_J$ and $C_J$ for all temperatures as shown in Figures 5e-f (see SI S16 for further detail including Im($Z_{NS-J}$) fitting). Figure 5e shows opposing temperature dependences for $R_J$ and $R_{NS}$ indicating hopping and band-like transport, respectively, with $R_{NS}/R_J$ increasing with temperature. Figure 5f shows a relatively small change in the junction capacitance, $C_J$, over the temperature range meaning the changes in the Re($Z_{NS-J}$) spectrum are dominated by changes in $R_J$. We find typical $C_J$ values of 6-8 fF, considerably smaller than a quantum capacitance (typically ~pF/mm$^2$)[55] but consistent with a geometric capacitance where $C_J = \varepsilon_r \varepsilon_0 A_J / l_J$. Taking $\varepsilon_r = 1$ and an inter-sheet distance of $l_J$ = 0.6 nm [Ref. [20]], gives a junction area of $A_J$ = 0.4-0.5 μm$^2$, in excellent agreement with SEM measurements which yield $\langle A_J \rangle$=0.4 μm$^2$ (Fig. 5f, inset, and SI S21).

To separately assess the transport mechanisms associated with the nanosheet and the junction, we examine the temperature dependence of $\rho_{NS}$, and $R_J$ in Figures 5g-h. Figure 5g shows $\rho_{NS}$ to scale as a power law, ($\rho_{NS} \propto T^{\alpha}$) with α ≈ 1.1, consistent with measurements on individual $MoS_2$ nanosheets (typically $\alpha$=0.5-1.9)[56-58]. This behavior implies band-like transport, limited by phonon scattering,[59] which is commonly seen for individual $MoS_2$ nanosheets with high carrier densities [refs. [58,60,61]], and is also in agreement with the THz spectroscopy data (triangles).

As these networks are junction-limited, the temperature dependence of $R_J$ in Figure 5h is similar to that of $\rho_{Net}$ showing the same the transition from VRH to activated behaviour. We propose this behaviour is consistent with Miller-Abrahams-type[53] hopping between nanosheets such that

$$R_J \approx R_{J0} \exp(2l_J / a) \exp(E_a / kT) \qquad (5a)$$

where $a$ is the localisation length. In SI S22, we derive an alternative version of 3D-VRH considering inter-nanosheet hopping from the conduction band-edge of one nanosheet to the conduction band of another yielding:

$$R_J \approx R_{J0} \exp\left(\frac{2l_J}{a}\right) \exp\left(\left[\frac{T_0}{T}\right]^{1/4}\right) \tag{5b}$$

Where the constant $T_0$ is here given by $T_0 \sim 76\pi\hbar^2 d_0 / ka^3 m$ with $d_0$ being the monolayer thickness. Fitting the data in Figure 5h yields $E_a$ = 55±2 meV and $T_0$ = (471±37)×$10^3$ K, values which are solely associated with the junction. Our $E_a$ value is smaller than other reported values (in the absence of gating[54]) which is consistent with our low $R_J$ (equation (5a)) and relatively high network carrier mobility.[62] Combining $T_0$ with $m = 0.7m_e$ and $d_0$ = 0.6 nm, we calculate $a$=0.7 nm, similar to published values for $MoS_2$ (0.2-3 nm).[54,63-65] The most probable hopping distance was ~ 2 nm, again consistent with inter-sheet hopping (see SI S22).

In summary, our simple model for conduction in nanoparticle networks is extremely powerful in allowing junction and particle resistance to be extracted from experimental data and demonstrates that the network properties such as conductivity can be tuned by adjusting the material dimensions. It represents an important new tool for analysis of printed networks of technological important nanomaterials.

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**Figures:**

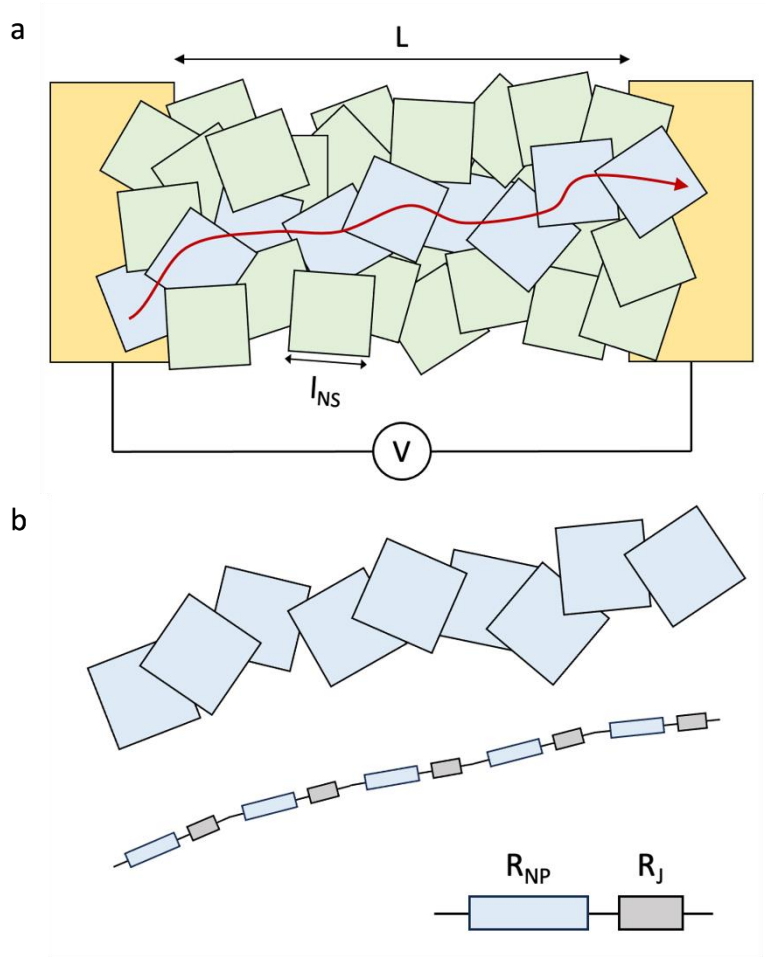


Figure 1: **Model Schematics**. **a**) Schematic illustrating a nanosheet network connected to two electrodes with channel length, $L$. The nanosheet lateral size is $l_{NS}$. While this schematic depicts a nanosheet network, a similar diagram could easily be produced to represent nanowire network. **b**) A single conducting path consisting of a linear array of nanosheets spanning the entire channel length. This conducting path can be considered as a chain of resistor pairs, with each pair consisting of a resistance representing a nanosheet and one representing the inter-sheet junction.

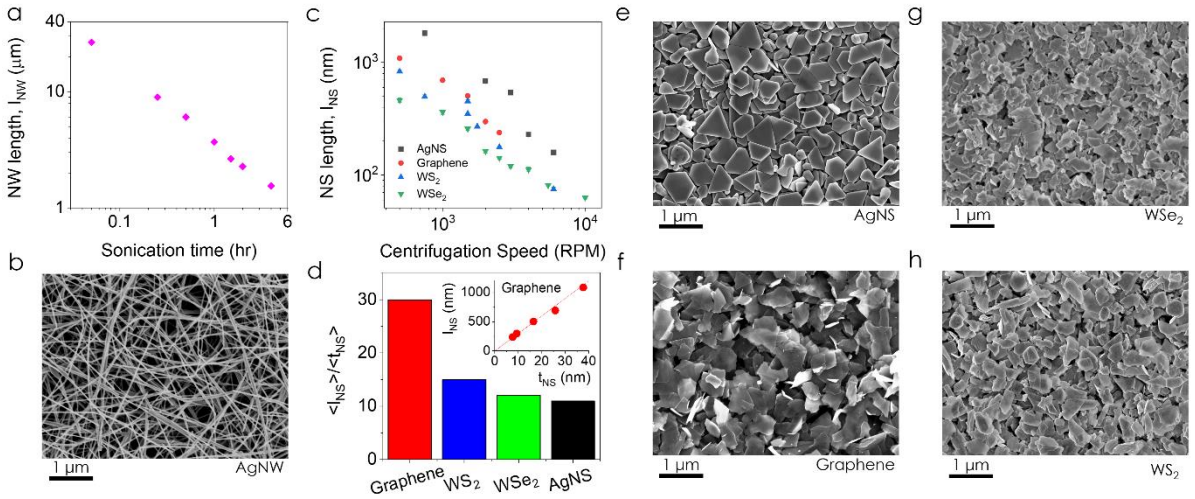


Figure 2: **Microscopic analysis of nanoparticles and networks**. **a**) Mean length of AgNWs size-selected by sonication-induced scission as a function of sonication time. **b**) SEM image of the top surface of a spray-cast network of AgNWs. **c**) Mean sizes of centrifuge-fractionated 2D materials: Ag, graphene, $WS_2$ and $WSe_2$ nanosheets, plotted versus centrifugation speed. **d**) Mean aspect ratio of all four 2D materials, averaged over all fractions. Inset: Exemplary data showing linear scaling between nanosheet length and thickness over five size-selected fractions of LPE graphene, consistent with aspect ratio of 30 (see SI S1 for all data). SEM images of the top surface of spray-cast networks of Ag (e), graphene (f), $WSe_2$ (g), $WS_2$ (h) nanosheets.

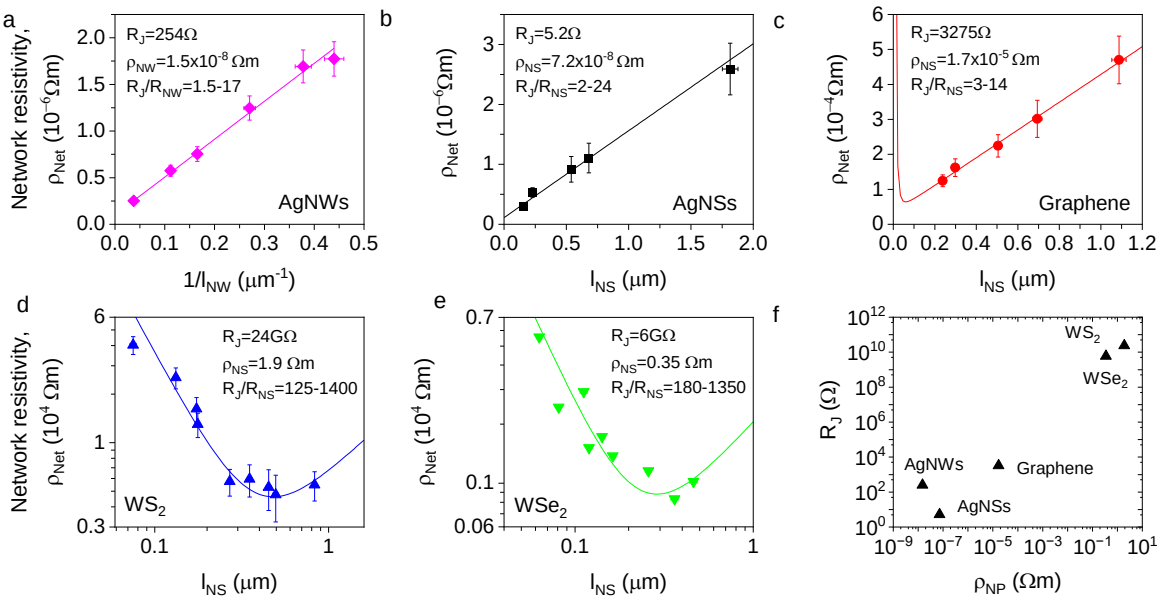


Figure 3: **Dependence of network resistivity on nanoparticle dimensions**. **a**) Resistivity of spray-cast silver nanowire networks versus inverse nanowire length. The line is a fit to equation (2a). **b-e)** Resistivity of spray cast nanosheet networks versus nanosheet length for networks of: AgNS (b), graphene (c), $WS_2$ (d), $WSe_2$ (e). In (b)-(e), the lines represent fits to equation (2b). In (a), (b), and (c), the carrier density is large allowing us to neglect the second square-bracketed term in the fitting equations. The behaviour in (b) and (c) is counter-intuitive as the general expectation is that smaller nanosheets lead to higher resistivity. Fitting yields values of $R_J$ and $\rho_{NP}$ (see SI S3). The latter parameter, combined with nanoparticles dimension's yields $R_{NP}$. Values of $R_J$ and $\rho_{NP}$ as well as ranges of $R_J/R_{NP}$ are given in the panels. **f**) Junction resistance, $R_J$, plotted versus nanoparticle resistivity, $\rho_{NP}$, demonstrating scaling.

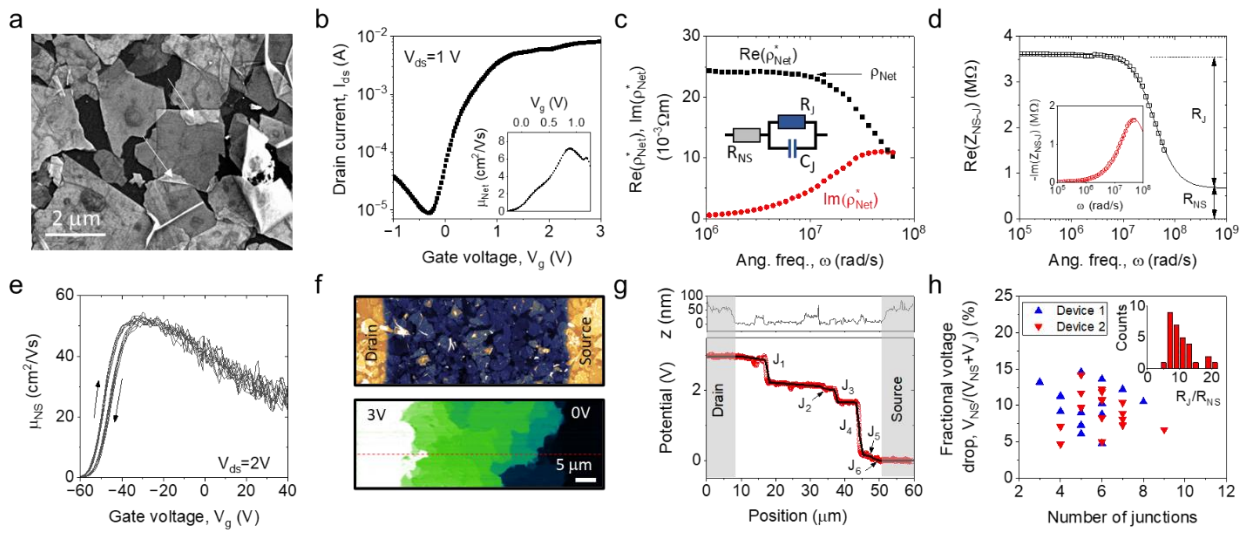


Figure 4: **Identification of nanosheet and junction resistances. a**) An SEM image of the surface of a network of electrochemically exfoliated (EE) $MoS_2$ nanosheets. The arrows point to two well-defined junctions. **b**) Field effect transfer curve for an electrolytically gated EE $MoS_2$ network very similar to that in (a). Inset: Gate-voltage-dependent mobility for an EE $MoS_2$ network nanosheet. Averaging over four such devices yields a mean (peak) mobility of $\mu_{Net}$=6.6±0.6 cm$^2$/Vs. **c**) Real and imaginary parts of the complex resistivity, both plotted as a function of angular frequency, ω, measured for a network of electrochemically exfoliated EE $MoS_2$ nanosheets similar to that in (a). Inset: The circuit element representing a nanosheet-junction pair. Here, $R_{NS}$ is the nanosheet resistance while $R_J$ and $C_J$ are the junction resistance and capacitance respectively. **d**) The real part of the impedance of a nanosheet-junction pair (Re($Z_{NS-J}$)) plotted versus ω. The data has been fitted using equation (4). The relationship between this curve and the junction and nanosheet resistances is shown by the arrows. Inset: -Im($Z_{NS-J}$) plotted as a function of ω. The solid line is a fit, see SI S11 for equation and fit parameters. **e**) Gate voltage dependent mobility for a representative individual EE $MoS_2$ nanosheet. **f**) Topographic AFM image (top) and *in-operando* KPFM image (bottom) of a section of a channel comprising an EE $MoS_2$ network. **g**) Topographic line profile (top) and potential profile (bottom) associated with the red dashed line in (f). In this section of channel, 6 sharp drops associated with inter-sheet junctions can be seen, labeled as $J_1$ to $J_6$. The flat regions represent the gradual drop of potential across nanosheets. The black line represents fits to the linear regions. **h**) Fraction of voltage dropped across nanosheets in a given section of channel plotted as a function of the number of junctions observed in that section. Inset: Histogram of $R_J/R_{NS}$ values as calculated from the fractional voltage drops in the main panel.

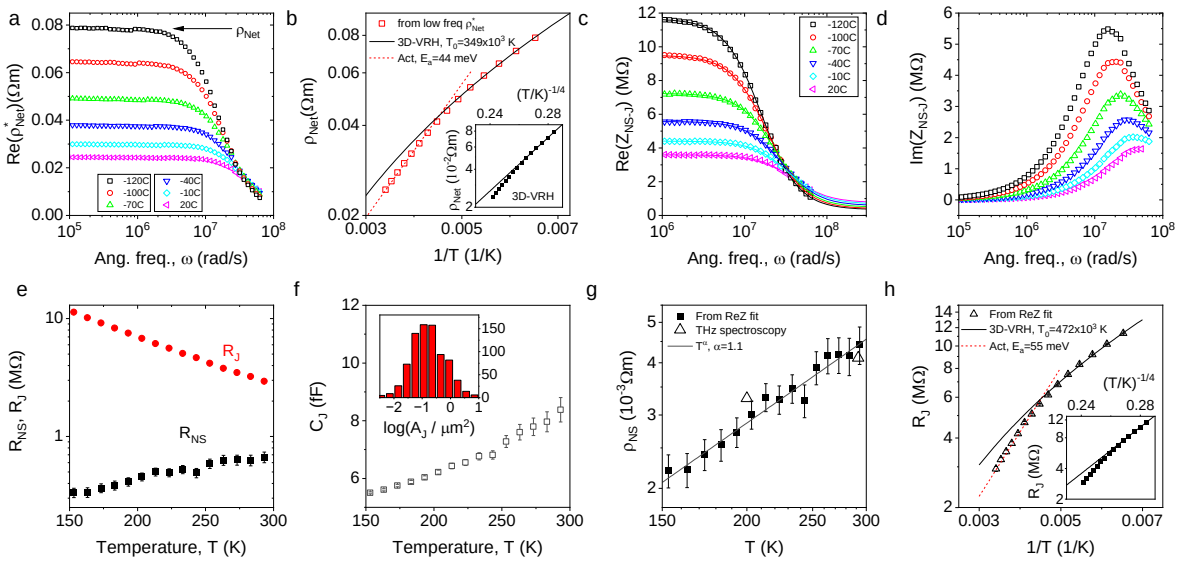


Figure 5: **Measurement of network, nanosheet, and junction transport using impedance spectroscopy**. **a**) Real part of the complex A.C. resistivity plotted as a function of angular frequency, ω, measured for a network of electrochemically exfoliated EE $MoS_2$ nanosheets at a range of temperatures. The arrow indicates that the D.C. network resistivity was found from $\rho_{Net} = \mathrm{Re}(\rho^*_{Net})_{\omega\to 0}$. **b**) D.C. network resistivity plotted as a function of 1/T and $T^{-1/4}$ (inset). The solid line is an activated fit while the dashed line is a fit to the 3D-VRH model. Fit parameters are given in the panel. **c-d**) Real (c) and imaginary (d) parts of the impedance spectrum of a single (average) nanosheet junction pair, measured at various temperatures. The curves in (c) are fitted using equation (4). **e**) $R_{NS}$ and $R_J$, extracted from fits to $\mathrm{Re}(Z_{NS\text{-}J}(\omega))$ spectra. **f**) Junction capacitance plotted versus temperature. Inset: Junction area histogram, measured from SEM images (plotted as $\log(A_J/\mu m^2)$). This distribution showed $\langle A_J \rangle$=0.41 $\mu m^2$. **g**) Resistivity of an (average) individual nanosheet, extracted from $R_{NS}$ ($\rho_{NS} \approx 2R_{NS}t_{NS}$), plotted as function of temperature. The solid line is a power law with exponent 1.1. The triangles represent the THz mobility of the nanosheets converted into resistivity using the measured carrier density of $3.8\times10^{23}$ $m^{-3}$. **h**) Junction resistance, $R_J$, plotted as a function of 1/T and $T^{-1/4}$ (inset). The solid line is an activated-behaviour fit while the dashed line is a fit to the 3D-VRH model. Fit parameters are given in the panel.

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